# Amlan Jyoti Sarmah

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Indian Citizen | Bhopal, Madhya Pradesh

## Education & Certificates

- B.Tech in Computer Science and Engineering from VIT Bhopal University CGPA: 8.58 / 10.0

## Work History/Projects/Internships

*Technical Team Lead, North East Club, VIT Bhopal* *Aug 2023 to Aug 2024*

- Led the technical team, overseeing development decisions to grow the club's digital presence.
- React.js and Tailwind CSS were used to build the club's official responsive website, improving student engagement and information accessibility.
- JavaScript and REST APIs were integrated for dynamic content updates, streamlining event announcements and reducing manual management overhead.
- Git and GitHub were used for version control, enabling smooth collaboration across a team of designers and developers for 5+ club events.

*Full-Stack Developer, Swupel e.u., Vienna, Austria (Early-stage startup, discontinued) Jan 2024 to Jun 2024*

- Joined as the second team member of an Austria-based early-stage startup, contributing across frontend and backend development.
- JavaScript and Chrome Extension APIs were used to build a browser extension following Manifest V3 standards, ensuring compliance and forward compatibility.
- Next.js and Tailwind CSS were used to build a marketing landing page for a Satellite API product, delivering a fast, SEO-friendly page to drive developer sign-ups.
- C++ was used to develop blockchain-related components, working on the core infrastructure layer of the product.

## Projects

*Custom C++ ML Library* *ml-from-scratch-cpp.vercel.app* | *GitHub*

- Implemented a lightweight, dependency-free ML library in C++ covering Linear Regression, Logistic Regression, and Naive Bayes with a modular class-based architecture.
- A TypeScript API exposes model training and inference endpoints, interfacing directly with the custom C++ models.
- A Python benchmark script compares performance against scikit-learn; results are visualised on a React frontend deployed on Vercel.

*Smart Tutor — Adaptive Learning Application* *adaptive-learning-application.vercel.app* | *GitHub*

- Built an adaptive tutoring system that adjusts question difficulty in real time based on individual student performance.
- Scikit-learn was used to train a performance classifier; Flask deployed it as a REST API for difficulty predictions.
- TypeScript, Express, and Zod handled authentication, question recommendation, and request validation in a separate backend service.
- MongoDB stores student profiles, performance records, and question banks to support the recommendation engine.

*songRoom — Real-Time Music Streaming & Chat* *github.com/AmlanJSarmah/songRoom*

- Built a real-time collaborative app allowing multiple users to stream songs and chat simultaneously in shared rooms.
- React.js was used on the frontend for a responsive, component-driven UI with live state updates.
- Node.js, Express, and Socket.io powered the backend and WebSocket layer for synchronised playback and low-latency chat.
- MongoDB was used to store user data, room metadata, and song history.